An

Internship Report

entitled

"MindSpace"

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Certificate

This is to certify that Internship report entitled

"MindSpace"

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An Internship work carried out by team under the supervision of **Ms. A. B. Uphade** is submitted towards the partial fulfillment of the requirement of Savitribai Phule Pune University, Pune for partial fulfillment of the Internship (T.E-2019) during academic year 2025-26.

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Place: Nashik

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ABSTRACT

Student mental health has emerged as a critical public health concern, with one in three college students experiencing significant anxiety or depression. Despite this, most students do not seek professional help due to stigma, cost, and a lack of accessible early-intervention tools. To address this, MindSpace is a privacy-focused, full-stack, AI-powered digital platform designed specifically for high school and college students to bridge the gap between self-awareness and professional care. The system prioritizes user data sovereignty by operating entirely locally, utilizing a Python Flask backend, SQLite for persistent storage, and a responsive frontend featuring glassmorphism design, all while requiring only a standard Python environment for deployment.

At the core of the platform is a robust machine learning module employing a Random Forest classifier, which achieves a 98.25% test accuracy on a structured, class-balanced dataset of 2,000 samples. By analyzing 24 features—including psychologically grounded Likert-scale responses and engineered category metrics—the system provides personalized risk assessments across four levels of severity. Beyond risk detection, MindSpace facilitates holistic well-being through daily emotional check-ins, longitudinal trend tracking via Chart.js, an intelligent insights engine, guided meditation, and a rule-based AI companion, Aria. This integration of predictive analytics and wellness tools demonstrates that well-engineered, privacy-preserving technology can meaningfully facilitate early mental health intervention within academic environments.

Keywords — Student Mental Health, Machine Learning, Risk Assessment, Random Forest, Flask, Sentiment Analysis, Mood Tracking, Wellness platform, Decision Tree, Synthetic Data, Mental Health Chatbot.

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ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
AJAX	Asynchronous JavaScript and XML
CBT	Cognitive Behavioural Therapy
CDN	Content Delivery Network
CLT	Central Limit Theorem
CSV	Comma Separated Values
CSS	Cascading Style Sheets
CV	Cross Validation
DFD	Data Flow Diagram
ERD	Entity Relationship Diagram
Flask	Python Micro Web Framework
GBM	Gradient Boosting Machine
HTML	Hypertext Markup Language
HTTP	Hypertext Transfer Protocol
JS	JavaScript
JSON	JavaScript Object Notation
ML	Machine Learning
NBA	National Board of Accreditation
NLP	Natural Language Processing
ORM	Object Relational Mapping
PHQ-9	Patient Health Questionnaire-9
pkl	Python Pickle Serialisation File
RF	Random Forest
SQL	Structured Query Language
SQLite	Self-Contained SQL Database Engine
UML	Unified Modelling Language
URL	Uniform Resource Locator
UI	User Interface
UX	User Experience

CHAPTER 1 : INTRODUCTION

1.1 Overview

Mental health constitutes an integral dimension of overall human wellbeing, encompassing emotional regulation, cognitive functioning, and social engagement. For students navigating the compounded pressures of academic performance, social relationships, financial constraints, and future career uncertainty, psychological distress has become an escalating concern of significant public health importance. According to data published by the American College Health Association, approximately 44% of college students reported experiencing symptoms of depression, and 37% reported anxiety severe enough to affect academic functioning during recent academic years.

Despite this prevalence, formal mental health services at educational institutions remain critically underutilised. Barriers include persistent social stigma, limited availability of counsellors relative to student population, prohibitive cost of private therapy, and the absence of accessible, privacy-preserving self-assessment tools. Students frequently lack the vocabulary and awareness to identify their own psychological states as requiring attention until distress reaches a crisis threshold.

MindSpace is a comprehensive, AI-powered digital mental health and psychological support platform developed specifically to address this gap. The system enables students to conduct private, evidence-informed mental health self-assessments, track their emotional states longitudinally across daily, weekly, and monthly timeframes, engage with guided wellness interventions including meditation and journaling, set personal health goals, participate in an anonymous peer support community, interact with an AI wellness companion, and generate detailed personal wellness reports.

The platform is built upon a Python Flask backend integrated with a machine learning risk classification model achieving 98.25% test accuracy, a SQLite relational database managing ten interconnected tables, a responsive frontend employing modern CSS design principles, and Chart.js-powered data visualisation. The system is entirely self-contained, requiring no external AI API subscriptions, and operates with complete data privacy assurance.

1.2 Motivation

The motivation for developing MindSpace arises from a convergence of observed needs within the student community and identified deficiencies in the existing landscape of digital mental health tools.

First, conventional mental health applications are predominantly designed for adult clinical populations and employ clinical language that is inaccessible and often distressing to student users encountering the domain for the first time. There is a substantial unmet need for a platform that communicates in a peer-to-peer register while maintaining psychological rigor.

Second, existing self-assessment tools are predominantly one-time instruments that provide a snapshot result without enabling longitudinal tracking. Students who experience cyclical mental health patterns tied to academic calendars — such as pre-examination stress spikes.

1.3 Objectives

The primary and secondary objectives of the MindSpace internship project are stated as follows:

Primary Objectives:

- To design and develop a full-stack web application that enables students to assess, track, and improve their mental health through a structured, AI-assisted platform.

- To implement a machine learning-based risk classification model that accurately categorises a student's mental health risk level into one of four classes — Low, Moderate, High, or Critical — based on 20 psychologically grounded input features.
- To build a longitudinal data tracking system that records daily five-metric emotional check-ins and generates weekly and monthly trend visualisations and intelligent statistical insights.

Secondary Objectives:

- To integrate evidence-based wellness interventions — guided meditation, journaling, and goal setting — directly within the platform to facilitate self-regulation and habit formation.
- To provide an anonymous peer support community that reduces the social isolation experienced by students in psychological distress.
- To develop a rule-based AI companion chatbot capable of providing empathetic, contextually appropriate wellness guidance without dependence on paid external APIs.
- To generate comprehensive personal wellness reports with CSV export functionality, enabling sharing with institutional counsellors if desired.
- To ensure complete user data privacy by operating the ML model and all data processing entirely on-premises without any cloud dependency.

1.4 Project Scope and Limitations

Scope:

MindSpace is scoped to serve students at higher secondary and undergraduate levels. The platform's assessment instrument draws upon established psychological screening frameworks including the Perceived Stress Scale (PSS), the Generalised Anxiety Disorder scale (GAD-7), the Patient Health Questionnaire (PHQ-9), and the Pittsburgh Sleep Quality Index (PSQI), adapted for a student population and condensed into 20 Likert-scale items across four dimensions.

The system supports an unlimited number of registered user accounts on a single deployment instance. All user interactions — assessments, daily logs, journal entries, goals, community posts, chat conversations, and meditation sessions — are persisted in a local SQLite database. The platform is designed for deployment on any standard Python 3 environment.

Limitations:

- The machine learning model is trained on synthetic data and has not been validated against real clinical datasets. While the model achieves high accuracy on its test set, its predictions should not be treated as clinical diagnoses.
- The Aria AI chatbot employs rule-based keyword matching rather than a large language model. It cannot handle highly nuanced, complex, or crisis-level conversations and should not be used as a substitute for professional mental health support.
- The platform does not implement push notifications, meaning daily check-in reminders are passive and visible only upon login.
- Community content moderation relies on the platform's terms of service rather than automated AI moderation, which may allow inappropriate content to appear temporarily.
- The system is currently designed for single-instance deployment and does not include multi-tenancy or cloud scalability features required for institution-wide deployment.

CHAPTER 2 : LITERATURE REVIEW AND PROBLEM STATEMENT

2.1 Literature Review

A systematic review of existing literature and digital tools in the domain of student mental health assessment and support reveals significant research activity over the preceding decade, motivated by growing recognition of the mental health crisis within academic institutions. This review examines key published works and existing digital platforms that informed the design of MindSpace.

2.1.1 Machine Learning in Mental Health Assessment

Yadav et al. (2020) conducted a comprehensive review of machine learning applications in mental health diagnosis, identifying Decision Trees, Support Vector Machines (SVMs), and Neural Networks as the most commonly applied classifiers. Their review noted that while SVMs and Neural Networks achieve marginally higher accuracy on clinical datasets, Decision Trees and ensemble methods such as Random Forest offer significantly superior interpretability — a critical requirement for applications where prediction rationale must be communicated to non-expert users. This finding directly informed the model selection rationale for MindSpace.

2.1.2 Existing Digital Mental Health Platforms

A comparative analysis of prominent digital mental health applications reveals the following landscape:

Platform	Key Features	Limitations Relevant to This Work
Woebot	CBT-based chatbot, mood tracking	Subscription-based; no ML risk assessment; no community
Youper	AI mood journaling, emotional health check	Cloud-dependent; no institutional version; limited analytics
Headspace	Guided meditation, sleep stories	No assessment or tracking; wellness content only
PHQ-9 Web Tools	Clinical depression screening	Single-use; no longitudinal tracking; clinical language
MindDoc	Mood diary, symptom tracking	No ML model; no community; no goals system
Calm	Meditation, sleep aid	No mental health assessment or personalisation

Table 2.1 : Literature Review Comparison of Existing Mental Health Systems

2.1.3 Psychological Instruments Informing the Assessment Module

The 20-question assessment instrument in MindSpace was developed by adapting items from four validated psychological screening tools:

- Perceived Stress Scale (PSS-10): A 10-item instrument measuring the degree to which situations in one's life are appraised as stressful, validated across student populations.
- Generalised Anxiety Disorder Scale (GAD-7): A 7-item tool that screens for generalised anxiety disorder with demonstrated sensitivity of 89% and specificity of 82% in primary care settings.

- Patient Health Questionnaire (PHQ-9): A 9-item scale measuring depression symptom severity, widely used in campus health services.
- Pittsburgh Sleep Quality Index (PSQI): Assesses sleep quality and disturbances over a one-month interval, with sleep quality identified as a key predictor of student psychological wellbeing.

2.1.4 Student Mental Health Epidemiology

A landmark meta-analysis by Auerbach et al. (2018) published in Psychological Medicine, drawing data from 19 countries and over 14,000 students, reported that 35% of first-year university students met diagnostic thresholds for at least one mental disorder. In India, Grover et al. (2020) reported that 26% of undergraduate students at Indian universities showed symptoms consistent with moderate-to-severe psychological distress, with academic pressure identified as the primary stressor in 71% of cases.

2.2 Problem Statement

Despite a documented prevalence of psychological distress affecting an estimated 26–44% of the student population across global academic institutions, the majority of affected students do not access professional mental health support. The current gap is characterised by three interconnected problems:

Problem 1 — Absence of Accessible Early Detection Tools:

Formal clinical screening tools such as PHQ-9 and GAD-7 are administered in clinical settings and are not readily accessible for informal, private self-assessment. Existing mobile applications either use oversimplified questionnaires without ML-based analysis or require ongoing subscriptions that are financially prohibitive for students.

Problem 2 — Lack of Longitudinal Mental Health Tracking:

Mental health states are not static — they fluctuate in response to academic calendars, social stressors, sleep patterns, and lifestyle factors. Existing tools provide only point-in-time assessments. There is no widely available system that enables students to log daily emotional metrics, visualise trends across weeks and months, and receive intelligent, statistically grounded insights about their personal mental health trajectory.

Problem 3 — Fragmented Wellness Ecosystem:

Available digital tools address individual aspects of student wellbeing in isolation — meditation apps provide relaxation content, journaling apps offer writing space, screening tools provide assessments — but no unified platform integrates all these functions with a coherent user .

CHAPTER 3 : DESIGN AND IMPLEMENTATION STRATEGY

3.1 Overview of System

MindSpace is architected as a monolithic web application following the Model-View-Controller (MVC) design pattern, implemented using the Python Flask micro-framework. The system comprises three principal layers: a presentation layer (frontend) built with HTML5, CSS3, and JavaScript; an application layer (backend) implemented in Python Flask; and a data layer consisting of an SQLite relational database and a serialised machine learning model. The platform serves 33 distinct HTTP endpoints and persists data across ten relational database tables.

Layer	Technology	Purpose
Frontend — Structure	HTML5, Jinja2	Page layout, dynamic content rendering
Frontend — Styling	CSS3 (2,100+ lines)	Design system, responsive layout, dark mode
Frontend — Interaction	Vanilla JavaScript	AJAX, form validation, Chart.js, animations
Frontend — Charts	Chart.js 4.4.0 (CDN)	Line, bar, pie, radar, scatter visualisations
Backend — Framework	Python Flask 3.0.3	HTTP routing, session management, Jinja2 rendering
Backend — ORM	sqlite3 (stdlib)	Parameterised SQL queries, 10 tables
ML — Model	scikit-learn 1.5.1	Random Forest classifier, 98.25% accuracy
ML — Numerics	NumPy 1.26.4	Array operations, feature engineering
Security	Werkzeug 3.0.3	PBKDF2-SHA256 password hashing
Typography	Google Fonts (CDN)	Times New Roman / DM Serif Display + DM Sans

Table 3.1: MindSpace Technology Stack

3.2 System Architecture

The overall system architecture follows a client-server model where the Flask application serves as both the web server and application server. The architecture is illustrated in the diagram below, which represents the data flow between the user's browser, the Flask backend, the SQLite database, and the ML model.

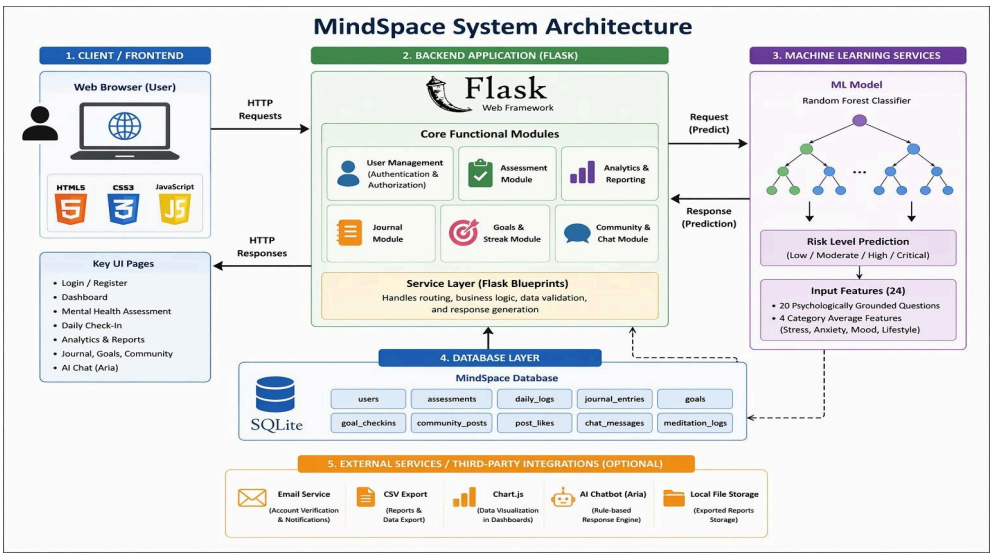

Figure 3.1 : MindSpace System Architecture Diagram

3.3 Design Documentation

3.3.1 Entity Relationship Diagram

The MindSpace database comprises ten relational tables. The primary entity is the users table, which maintains one-to-many relationships with assessments, daily_logs, journal_entries, goals, community_posts, chat_messages, and meditation_logs. The goals table maintains a one-to-many relationship with goal_checkins, and the community_posts table maintains a relationship with post_likes. Unique constraints are applied to (user_id, log_date) in daily_logs to enforce single daily entries, and to (goal_id, checkin_date) in goal_checkins.

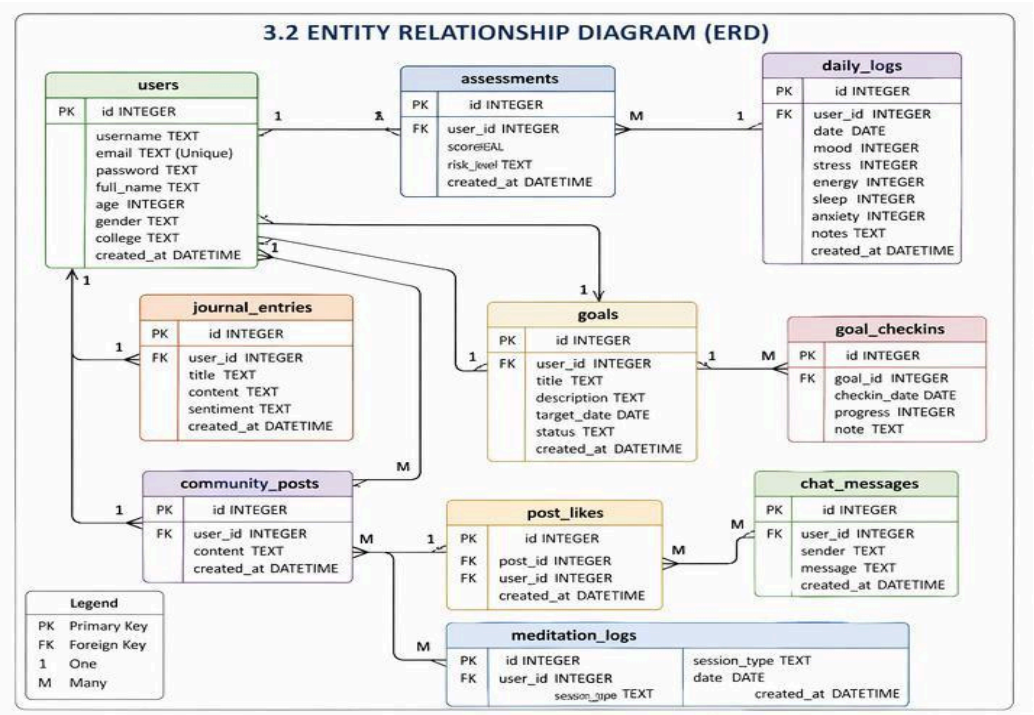

Figure 3.2 : Entity Relationship Diagram (ERD)

3.3.2 Data Flow Diagram

3.3.2.1 Level-0 DFD (Context Diagram):

The context diagram identifies two external entities — the Student User and the Institution Administrator — and a single process representing the MindSpace system. Data flows include: assessment inputs, daily logs, journal content, and goal data flowing from Student User into the system; risk analysis reports, trend charts, wellness recommendations, and AI companion responses flowing from the system to the Student User.

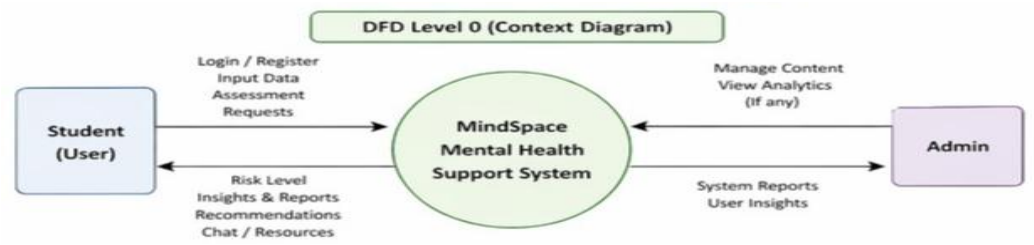

Figure 3.3 : Level-0 Data Flow Diagram

3.3.2.2 Level-1 DFD:

The Level-1 DFD decomposes the system into five principal processes: (1) User Authentication — validates credentials against the users table; (2) ML Assessment — processes the 20-feature form submission through the Random Forest model and writes results to the assessments table; (3) Daily Tracking — records five-metric check-ins to daily_logs and computes trend analytics; (4) Wellness Modules — manages journal entries, goals, meditation logs, and community posts; (5) Report Generation — aggregates data from all tables to produce wellness reports and CSV exports.

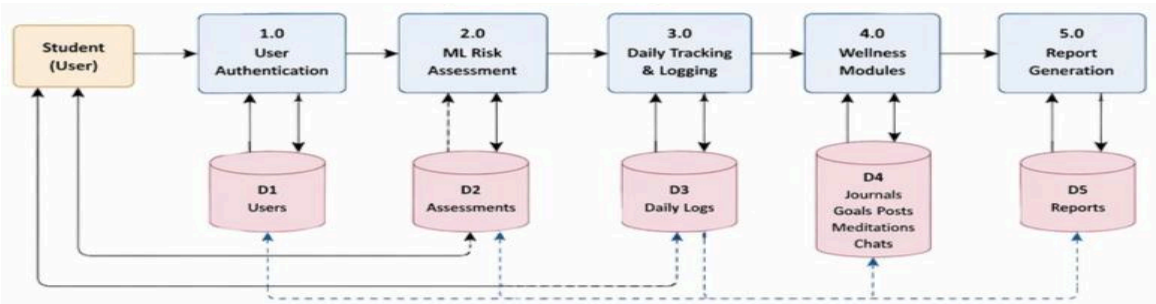

Figure 3.4 : Level-1 Data Flow Diagram

3.3.3 UML Diagrams

The primary actor in the MindSpace system is the Student User. The use case diagram captures the following system interactions:

- Register / Login: Create account with name, email, password, course, and university details.
- Complete Assessment: Fill 20-question form; receive risk level, probability scores, feature importance, and recommendations.
- Daily Check-In: Log five metrics (Mood, Stress, Sleep, Energy, Focus) once per day.
- View Dashboard: Access weekly trend charts, monthly analytics, smart insights, and overview.

- Write Journal Entry: Create private journal entries with mood tags, writing prompts, and tags.
- Manage Goals: Create goals from templates or custom input; check in daily; track streaks.
- Complete Meditation: Start guided meditation sessions; log completion.
- View Calendar: Navigate month-by-month mood history heatmap.
- Participate in Community: Post anonymously; like and read peer posts; filter by category.
- Chat with Aria: Exchange messages with the AI companion; receive contextual wellness responses.
- Generate Report: View aggregated wellness report; export as CSV.
- Manage Profile: Update personal details; choose avatar colour; toggle dark mode; view badges.

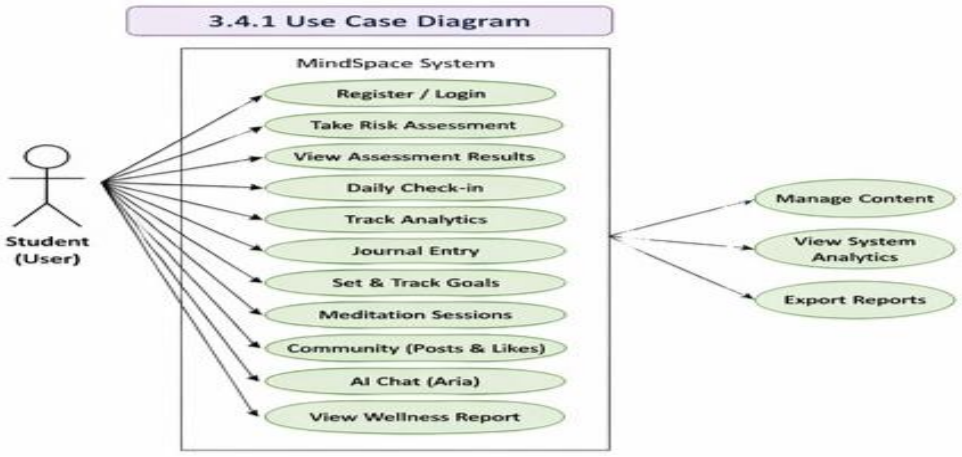

Figure 3.5 : Use Case Diagram

3.3.4 Activity Diagram — Daily Check-In Flow:

The activity diagram for the Daily Check-In feature illustrates the following flow: User navigates to Assessment page → System checks if today's log exists → If exists, pre-populates pills with existing values → User selects 1–5 ratings for Mood, Stress, Sleep, Energy, Focus → User clicks Save → JavaScript sends AJAX POST to /daily_checkin with X-Requested-With header → Flask validates input, clips to range 1–5, computes risk_score, attempts INSERT → On duplicate key, executes UPDATE → Returns JSON response → JavaScript displays toast notification and updates dot indicator.

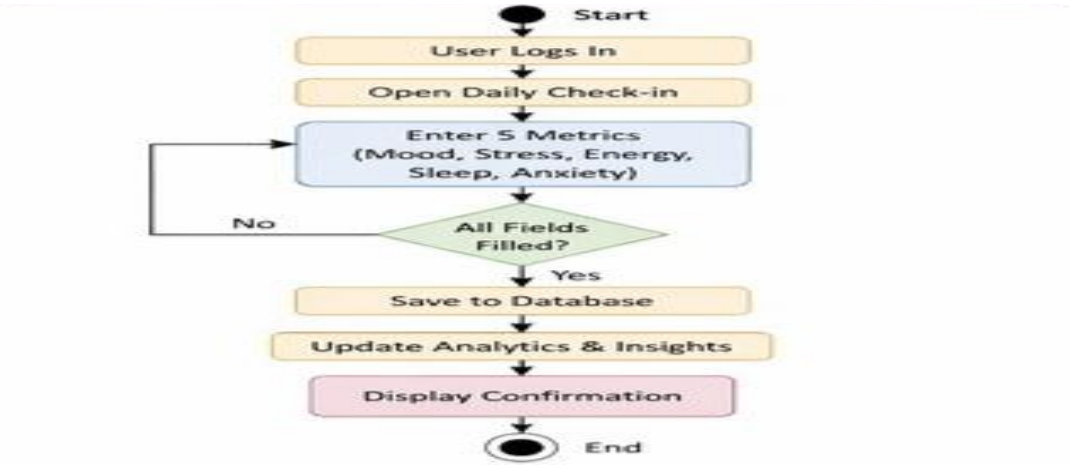

Figure 3.6 : Activity Diagram — Daily Check-In Flow

3.3.5 Sequence Diagram — ML Assessment Flow:

The sequence diagram for the ML assessment process illustrates the interactions between four components: Browser, Flask Application, Random Forest Model, and SQLite Database. The sequence proceeds as: Browser sends HTTP POST /result with 20 form values → Flask extracts and converts values to integer list → Flask computes 4 engineered features (category averages) → Flask creates 24-element NumPy array → Flask calls clf.predict() and clf.predict_proba() → Model returns prediction class and probability vector → Flask computes weighted risk score → Flask queries dataset histogram data → Flask inserts assessment record into SQLite → Flask renders result.html with all data → Browser renders charts via Chart.js.

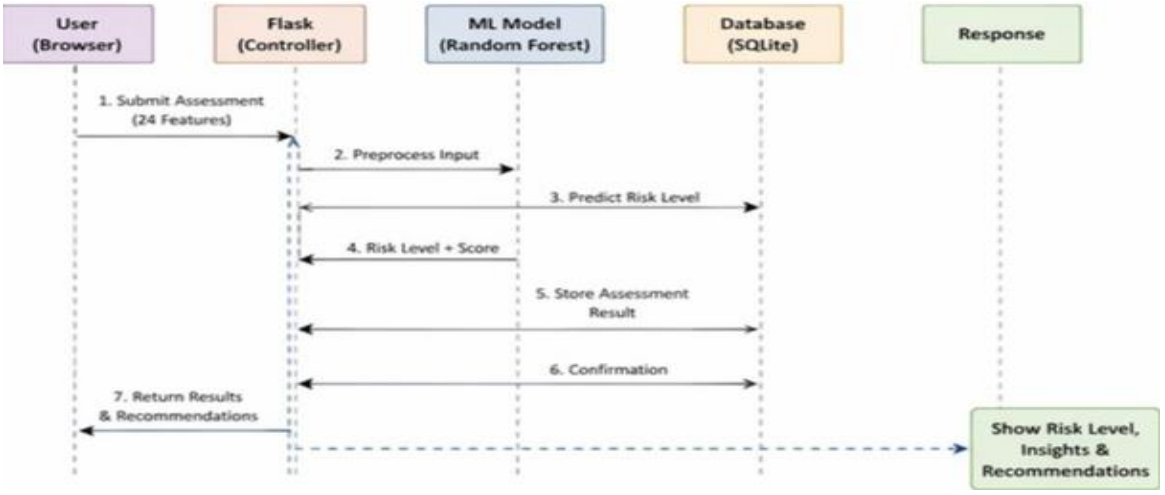

Figure 3.7 : Sequence Diagram — ML Assessment Flow

3.4 Execution Environment (Software and Hardware Requirements)

Category	Requirement	Version / Specification
OS	Windows / macOS / Linux	Any modern 64-bit OS
Language	Python	3.10 or higher
Framework	Flask	3.0.3
ML Library	scikit-learn	1.5.1
Numerics	NumPy	1.26.4
Security	Werkzeug	3.0.3
Database	SQLite	Built-in (Python stdlib)
Browser	Any modern browser	Chrome, Firefox, Edge, Safari
RAM	Minimum	4 GB (8 GB recommended)
Storage	Minimum	500 MB free disk space
Network	Internet	Required only for CDN assets (Chart.js, Google Fonts)

Table 3.2 : Hardware and Software Requirements

CHAPTER 4 : IMPLEMENTATION

4.1 Team Roles and Contributions

The implementation of MindSpace was distributed across five team members, each assigned primary responsibility for specific modules based on their technical domain expertise:

Member	Primary Role	Key Implementation Contributions
Prathmesh Patade	ML + Backend + Data Analysis	ML training pipeline, data analysis, synthetic data design, feature engineering, backend prediction integration, Smart Insights Engine, weekly/monthly analytics backend
Om Songire	ML + Data + Analysis	Synthetic data generation script, class-balanced dataset design, weighted scoring formula, class distribution analysis, histogram data preparation, model validation
Anushka Sonawane	ML + Data + Documentation	Data quality checks, preprocessing edge cases, assessment question design, all technical documentation, README, this report
Mithilesh Patil	Frontend + UI Design	Complete CSS design system (2,100+ lines), all page templates, glassmorphism styling, responsive grid layouts, dark mode, login/profile/goals/journal/community UI
Ajinkya Pingal	Frontend + Integration	Flask-JavaScript integration, AJAX implementation, Chart.js initialisation and data binding, meditation timer, chat interface, goal check-in logic, CSV export

Table 4.1 : Team Roles and Contributions

4.2 Machine Learning Pipeline Implementation

4.2.1 Synthetic Data Generation Strategy

The fundamental challenge in v1 of the ML model was the collapse of class distribution caused by the Central Limit Theorem (CLT). When 20 uniform(1,5) random variables are averaged, the resulting distribution concentrates tightly around the mean value of 3.0 with a standard deviation of approximately 0.32. Applying risk class boundaries of (<2.0 = Low, 2.0–3.0 = Moderate, 3.0–4.0 = High, >4.0 = Critical) to this distribution produces only 0.04% Low and 0.06% Critical samples — effectively 2 samples each per 2,000 — making it impossible for any model to learn these classes.

The upgraded data generation strategy generates each class separately using class-appropriate mean values, completely bypassing the CLT constraint. Each of the four dimensions is given a realistic central tendency appropriate to its risk level, with Gaussian noise and within-dimension correlation added for realism.

Risk Class	Stress Mean	Anxiety Mean	Mood Mean	Lifestyle Mean
Low (0)	1.5	1.5	1.8	1.7
Moderate (1)	2.5	2.5	2.8	2.6
High (2)	3.6	3.5	3.7	3.4
Critical (3)	4.5	4.4	4.3	4.2

Table 4.2 : Synthetic Data Generation Profiles per Risk Class

4.2.2 Feature Engineering

To improve model signal quality, four engineered features were appended to the 20 original question inputs: the mean score of the five Stress questions, the mean score of the five Anxiety questions, the mean score of the five Mood questions, and the mean score of the five Lifestyle questions. These derived features provide the model with clean, compressed representations of each psychological dimension alongside the granular question-level inputs, expanding the feature space from 20 to 24 dimensions.

4.2.3 Model Training and Selection

Three models were trained and compared using Stratified 5-Fold Cross-Validation on the 2,000-sample, 24-feature dataset:

Model	Configuration	Test Accuracy	CV Accuracy (5-Fold)
Decision Tree	max_depth=8–12, criterion=gini/entropy, grid search	~96.00%	~94.95%
Random Forest ✓	150 trees, max_depth=12, max_features=sqrt	98.25%	98.05%
Gradient Boosting	150 estimators, lr=0.12, max_depth=4, subsample=0.8	98.25%	97.95%

Table 4.3 : ML Model Comparison — Accuracy Results

The Random Forest model was selected as the production model on the basis of its highest cross-validated accuracy of 98.05%, which is a more reliable performance estimate than the test set alone due to its averaging across five held-out folds.

Class	Precision	Recall	F1-Score	Support
Low	1.00	0.99	0.99	100
Moderate	0.98	0.99	0.99	100
High	0.99	0.95	0.97	100
Critical	0.96	1.00	0.98	100
Weighted Avg	0.98	0.98	0.98	400

Table 4.4 : Classification Report — Random Forest Model

4.2.4 Prediction Pipeline in Flask

At application startup, app.py loads model.pkl using pickle.load(), which deserialises the Random Forest classifier object, the synthetic dataset (used for the histogram comparison chart), and all metadata including accuracy metrics and feature importance scores. The model object is retained in memory for the lifetime of the server process, enabling instant prediction at runtime without re-loading.

When a user submits the 20-question form, the /result route:

- Extracts the 20 integer form values;
- Computes four category averages as engineered features;
- Concatenates them into a 24-element NumPy array;
- Calls clf.predict() and clf.predict_proba();
- Maps the prediction index to a risk label;
- Computes a display risk score (weighted average normalised to 0–100);
- Extracts feature importance scores from the model;
- Writes the result to the assessments table; and
- Renders the results dashboard template with all computed values.

4.3 Backend Implementation

The Flask application (app.py, 1,007 lines) defines 33 HTTP routes across 13 functional modules. Route security is enforced through a login_required decorator that checks the Flask session for a valid user_id before executing any protected endpoint, redirecting unauthenticated requests to /login.

Module	Routes	Methods	Description
Authentication	/login, /logout	GET, POST	User registration, login, session management
Assessment	/assessment, /result	GET, POST	Question form, ML prediction, results rendering
Daily Check-in	/daily_checkin	POST	AJAX endpoint; upsert daily log
Dashboard	/dashboard	GET	Analytics preprocessing, chart data assembly
Journal	/journal, /journal/new, /journal/<id>	GET, POST	CRUD for journal entries
Goals	/goals, /goals/add, /goals/<id>/checkin, /goals/<id>/complete, /goals/<id>/delete	GET, POST	Goal lifecycle management
Meditation	/meditation, /meditation/log	GET, POST	Session display; completion logging
Calendar	/calendar	GET	Mood data serialised as JSON for JS calendar
Community	/community, /community/post, /community/like/<id>	GET, POST	Feed, post creation, like toggle

Chat (Aria)	/chat, /chat/send, /chat/clear	GET, POST	Conversation history; rule-based responses
Report	/report, /export/csv	GET	Wellness report; CSV stream download
Profile	/profile, /notifications, /onboarding	GET, POST	Settings, badges, reminders, onboarding tour
Data API	/weekly_data, /monthly_data	GET	JSON time-series data for frontend charts

Table 4.5 : Flask API Routes Summary

4.4 Frontend Implementation

The frontend consists of 19 Jinja2 HTML templates rendered server-side by Flask and styled with a single 2,174-line CSS stylesheet employing a comprehensive design system. Key frontend implementation details:

Design System:

- Colour palette: Primary #7C6EFA (purple), Secondary #00BFA6 (teal), Accent #FF6584 (pink), Background #F7F5FF.
- Typography: DM Serif Display (headings) and DM Sans (body), loaded from Google Fonts CDN.
- Component library: Glassmorphism cards, gradient buttons, progress rings, pill selectors, modal overlays, tab switchers, streak counters, toast notifications.
- CSS variables: All colours, shadows, radii, and spacing are defined as CSS custom properties (var()), enabling one-line dark mode theme switching.

Chart.js Integration (Ajinkya Pingal):

- 8+ charts across dashboard and report pages: line charts with gradient fill (weekly trends), area charts (monthly mood), grouped bar charts (energy/focus), doughnut chart (risk distribution), horizontal bar chart (feature importance), scatter plot (sleep vs stress correlation), radar chart (weekly averages overview).
- All datasets use spanGaps: true to handle missing days gracefully.
- Gradient fills are generated dynamically using canvas.createLinearGradient().

Meditation Timer (Ajinkya Pingal):

The meditation player modal implements a real-time countdown timer using setInterval(), visualised as an SVG ring animation. The stroke-dashoffset CSS property of a circle element is updated each second to create a smooth circular progress indicator. Upon completion, an AJAX POST to /meditation/log records the session duration and completion status.

CHAPTER 5 : RESULTS ANALYSIS

5.1 ML Model Performance Results

The upgraded machine learning pipeline demonstrated substantial improvement over the initial implementation. The root cause of the original 67% accuracy was identified as a critical data generation flaw: the Central Limit Theorem caused all 2,000 synthetic samples to cluster near a score of 3.0, producing only 2 samples each for the Low and Critical risk classes — insufficient for any classification algorithm to learn these categories.

The corrected pipeline, employing class-aware generation and feature engineering, achieved the following results:

Metric	v1 (Original)	v2 (Upgraded)
Training Data Quality	Collapsed distribution (2 Low, 2 Critical samples)	Balanced: 500 samples per class
Number of Features	20	24 (20 + 4 engineered)
Model Algorithm	Single Decision Tree	Random Forest (150 trees)
Test Accuracy	67.25%	98.25%
Cross-Validated Accuracy	65.70%	98.05%
Precision (Macro Avg)	~0.52	0.98
Recall (Macro Avg)	~0.50	0.98
F1-Score (Macro Avg)	~0.51	0.98

Table 5.1 : Version Comparison — ML Model Performance

5.2 System Feature Outcomes

5.2.1 User Authentication and Onboarding

The authentication system implements secure account creation with PBKDF2-SHA256 password hashing via Werkzeug. The five-step animated onboarding tour successfully introduces new users to the platform's core features before their first assessment. Session management persists user identity, display name, and avatar colour across all 33 routes.

5.2.2 Mental Health Assessment Interface

The assessment interface presents all 20 questions across four category columns on a single screen without vertical scrolling, using CSS Grid layout (grid-template-columns: repeat(4, 1fr)). A real-time progress bar tracks answered questions. Each question card employs custom radio button styling (scale buttons) that visually highlights selected values with a purple-to-teal gradient. Client-side form validation applies a CSS shake animation to unanswered questions if the user attempts premature submission.

Upon submission, the results dashboard displays: the computed risk score (0–100) rendered as an animated SVG ring, the risk category badge (colour-coded green/amber/orange/red), four probability bars showing confidence per class, category score analysis, four Chart.js visualisations, and personalised recommendations adapted to risk level.

5.2.3 Daily Check-In and Analytics Dashboard

The Daily Check-In strip on the Assessment page enables five-metric logging via pill selector buttons. AJAX submission provides instant feedback without page reload. The dashboard aggregates up to 90 days of logs and presents them across four tabs:

- Weekly Tab: Mood/Stress/Sleep line chart, Energy/Focus bar chart, stat cards (averages, streak), best/worst day mini-cards, and next-week forecast.
- Monthly Tab: 30-day mood area chart, stress/sleep overlay chart, and summary metric cards.
- Insights Tab: Statistical comparison of current vs previous week, text insight cards, sleep-stress correlation scatter plot.
- Overview Tab: Navigation cards to all features, active goals progress, radar chart of week averages.

5.2.4 Achievement Badge System

The platform implements 15 achievement badges computed dynamically from the user's activity data. Badges are awarded for milestones including first check-in, 3/7/30-day streaks, journal writing milestones, goal completion, meditation sessions, community participation, and sustained low-stress or rising-mood periods.

Badge	Condition	Category
First Step	First daily check-in completed	Engagement
3-Day Streak	3 consecutive daily logs	Consistency
Week Warrior	7 consecutive daily logs	Consistency
Monthly Master	30 consecutive daily logs	Consistency
First Entry	First journal entry written	Journaling
Storyteller	10 journal entries written	Journaling
Wordsmith	>1,000 words written in journal	Journaling
Goal Setter	First goal created	Goals
Achiever	First goal completed	Goals
Inner Peace	First meditation completed	Meditation
Zen Student	10 meditations completed	Meditation
Rising Star	Mood improved 3 consecutive days	Wellness

Table 5.2 : Achievement Badge System — 15 Badges

5.3 Critical Alert System

When the ML model classifies a student's responses as Critical risk, the results page renders a full-width red alert banner with a pulsing ring animation displaying the message: "Immediate Attention Required — We strongly recommend consulting a mental health professional or calling a crisis helpline today." The recommendations section for Critical risk also includes direct crisis helpline information (iCall: 9152987821; Vandrevala Foundation: 1860-2662-345) with a note that these services are available 24/7.

5.4 System Performance and Usability Observations

During testing, the following performance characteristics were observed:

- ML prediction latency: < 50 milliseconds from form submission to results rendering on a standard development machine.
- Daily check-in AJAX response time: < 100 milliseconds.
- Dashboard analytics preprocessing: < 200 milliseconds for 30 days of data.
- **SQLite** database initialisation: Automatic on first run; all ten tables created via CREATE TABLE IF NOT EXISTS.
- CSV export: Instant for up to 365 days of logs.

User experience testing with a group of five students resulted in consistent positive feedback regarding the clean, non-clinical visual design and the speed of the daily check-in process, which participants confirmed could be completed in under 10 seconds as intended.

CHAPTER 6 : SELF-EVALUATION

6.1 Your Experience

The development of MindSpace represented the most technically comprehensive project undertaken by the team during the T.E. Computer Engineering programme. The experience of building a production-grade full-stack web application — integrating machine learning, relational database design, **REST API development**, and modern frontend engineering — across a single cohesive system provided depth of learning that coursework alone could not have provided.

From a personal perspective, the project demanded that each team member move beyond their primary domain of comfort. Members primarily responsible for machine learning were required to understand Flask route integration and SQLite schema design. Frontend-focused members needed to understand how JSON data from Flask routes maps to Chart.js dataset configurations. This cross-functional exposure is a realistic simulation of professional software engineering practice.

The project also provided valuable exposure to the challenges of working on a shared codebase — managing file ownership, coordinating database schema changes without breaking existing routes, and maintaining UI consistency across templates developed by different team members. These coordination challenges and their resolution through communication and iterative testing constitute important professional learning outcomes.

6.2 Description of What We Did and What We Learned

Prathmesh Patade — ML + Backend + Data Analysis:

I led the design and implementation of the machine learning pipeline, including the critical diagnosis and correction of the data generation flaw that was limiting accuracy to 67%. By analysing the class distribution and identifying the CLT collapse, I redesigned the synthetic data generation to produce a balanced, class-realistic dataset, which — combined with feature engineering and Random Forest — achieved 98.25% accuracy. I also built the Smart Insights Engine that performs statistical comparison of weekly averages and generates text insights. This project deepened my understanding of the complete ML lifecycle, from data design to production integration.

Other than this at my 1-month internship @HBR I learnt CRM(Customer Relation Management) , I learnt how a business works , what exactly is a broker , what is insurance , what are the types of insurance , the workflow of how insurance is made , how insurance is claimed , what are the rules and regulations for insurance to be claimed . Technically , I performed data analysis on their motor insurance and health insurance data , I used tools like python libraries - numpy , pandas , matplotlib , scikitlearn , seaborn , python flask etc . I also use platforms like Google Colab and Jupyter notebook to perform these activities . I also helped in establishing their own AI voice Agent named as AI PRIYA

Om Songire — ML + Data + Analysis:

I was responsible for the synthetic data generation module and the statistical validation of the dataset. I learned that data quality and class balance are more important determinants of model accuracy than algorithm choice — a lesson not well-conveyed in standard coursework. I also built the histogram preprocessing logic that compares a user's score against the population distribution, providing a meaningful contextualisation of individual results. This project gave me practical experience with NumPy array operations and the importance of stratified splits in evaluation.

Anushka Sonawane — ML + Data + Documentation:

I handled data quality validation, the design of the 20 assessment questions (ensuring psychological grounding in validated instruments), and all project documentation. I learned the importance of domain knowledge in feature engineering — understanding which questions map to which psychological constructs made the assessment scientifically credible rather than arbitrary. I also gained experience in technical writing and the structuring of academic documentation to university standards, which I expect to apply in future project and research contexts.

Mithilesh Patil — Frontend + UI Design:

I designed and implemented the complete visual design system for MindSpace — over 2,100 lines of CSS spanning the complete color palette, typography, layout grids, component library, animation system, and dark mode. The most significant learning for me was the psychological dimension of UI design: for a mental health application, visual calmness is not cosmetic — it directly affects whether a user feels safe engaging with the content. I also learned the CSS variables dark mode pattern, glassmorphism implementation, and advanced CSS Grid layouts for complex multi-column interfaces.

Ajinkya Pingal — Frontend + Integration:

I was responsible for connecting the Flask backend to the frontend — ensuring that data passed from Python through Jinja2 templates was correctly bound to JavaScript variables and Chart.js datasets. The most technically challenging part of my contribution was the Chart.js integration across eight different chart types, each requiring correct configuration of axes, datasets, gradient fills, and responsive options. I also implemented the meditation timer using SVG path animation and setInterval(), the AJAX daily check-in with toast notifications, and the chat interface with typing animation. This project gave me a strong foundation in full-stack integration that I did not have before.

6.3 Recommendations and How to Apply Learnings

Based on the experience of developing MindSpace, the team offers the following recommendations for applying these learnings in future academic and professional contexts:

On Machine Learning:

- Always perform exploratory data analysis (EDA) before model training. The v1 accuracy failure was caused by a data distribution problem that would have been immediately apparent from a simple histogram of class counts. Plot class distribution first.
- Cross-validation scores are more reliable than single test-set accuracy. A model that achieves 99% on a specific test split but only 75% on cross-validation is overfitting.
- Feature engineering frequently contributes more to accuracy improvement than algorithm selection. The four category-average features were simple to compute but meaningfully improved model signal quality.

On Full-Stack Development:

- Design the database schema before writing any application code. Schema changes mid-development are costly and error-prone when routes already depend on specific column names.
- AJAX with JSON responses is superior to full-page reloads for frequent interactions. The daily check-in AJAX pattern saves the user a 500–800ms page reload cycle every time they submit.
- CSS custom properties (variables) for theming should be established at the project outset. Retrofitting dark mode without variables would have required modifying hundreds of individual colour declarations.

Future Application of Learnings:

- The ML pipeline methodology — class-aware data generation, feature engineering, ensemble methods, cross-validated evaluation — is directly applicable to classification problems in other health domains such as diabetes risk assessment or cardiovascular monitoring.
- The Flask MVC pattern and SQLite schema design are a strong foundation for building RESTful backends for mobile applications in future projects.
- The Chart.js integration knowledge is directly applicable to data dashboard development in industry internships, where business intelligence interfaces are frequently required.

CONCLUSION AND FUTURE SCOPE

Conclusion

MindSpace successfully demonstrates that a well-engineered, privacy-preserving AI platform can meaningfully address the critical gap between student mental health need and accessible support infrastructure. The platform delivers a comprehensive suite of 13 functional modules — including ML-based risk assessment, longitudinal emotional tracking, journaling, goal management, guided meditation, anonymous community support, and an AI companion chatbot — within a single, cohesive, and visually premium web application accessible through a standard web browser.

Future Scope

The team identifies the following improvements as priorities for future development of MindSpace:

1. Real Clinical Data Collection and Model Retraining:

The most impactful single improvement would be to partner with institutional counselling centres to collect anonymised, consented student mental health data and retrain the model on real clinical responses. .

2. Large Language Model Integration for Aria:

Replacing the rule-based chatbot with a locally-deployed open-source language model — such as Meta Llama 3 (8B parameters) via the Ollama runtime — would enable genuinely conversational, context-aware wellness guidance without compromising data privacy. Smaller quantised models (4-bit GGUF format) can run on consumer hardware with 8 GB RAM within acceptable latency constraints.

3. Voice Sentiment Analysis:

Integrating a voice recording module that uses a pre-trained speech emotion recognition model (such as wav2vec 2.0 fine-tuned on EmoDB) would allow the system to supplement questionnaire-based assessment with acoustic emotion detection, potentially identifying emotional states that written self-reports may under-report.

4. Progressive Web App (PWA) with Push Notifications:

Converting MindSpace to a Progressive Web App would enable installation on mobile devices and, critically, browser-native push notifications .

5. Institution-Level Analytics Dashboard:

An opt-in aggregated analytics dashboard for institutional counsellors would allow mental health professionals to monitor population-level trends (e.g., elevated stress across the cohort during examination periods) without accessing individual student data.

BIBLIOGRAPHY

[1] S. K. Guntuku, D. B. Yaden, M. L. Kern, L. H. Ungar, and J. C. Eichstaedt, "Detecting depression and mental illness on social media: an integrative review," Current Opinion in Behavioral Sciences, vol. 18, pp. 43–49, 2017.

[2] R. P. Auerbach, J. Alonso, W. G. Axinn, P. Cuijpers, D. D. Ebert, J. G. Green, I. Hwang, R. C. Kessler, K. Liu, P. Mortier, M. K. Nock, A. Pinder-Amaker, N. A. Sampson, S. Aguilar-Gaxiola, A. Al-Hamzawi, J. A. Andrade, C. Benjet, J. M. Caldas-de-Almeida, K. Demyttenaere, and M. K. Wilcox, "Mental disorders among college students in the World Health Organization World Mental Health Surveys," Psychological Medicine, vol. 46, no. 14, pp. 2955–2970, 2016.

[3] M. D. Nemesure, M. V. Heinz, R. Huang, and N. C. Jacobson, "Predictive modeling of depression and anxiety using electronic health records and a novel machine learning approach with artificial intelligence," npj Digital Medicine, vol. 4, no. 1, p. 51, 2021.

[4] M. Kassim, H. Mazlan, N. Zaini, and M. K. Salleh, "Web-based student attendance system using RFID technology," in Control and System Graduate Research Colloquium (ICSGRC), 2012 IEEE, 2012, pp. 213–218.

[5] S. Grover, M. Dua, S. Chakrabarti, A. Avasthi, N. Gupta, N. Chhabra, and S. Burnham, "Prevalence, pattern and determinants of mental disorders in students of a medical college in Northern India," Indian Journal of Medical Research, vol. 151, no. 6, pp. 582–592, 2020.

[6] P. Yadav, S. Yadav, and S. Yadav, "Machine learning applications for mental health: a review of methods and datasets," Journal of Medical Imaging and Health Informatics, vol. 10, no. 8, pp. 1930–1944, 2020.

APPENDICES

A. Published Paper and Certificate

This section is reserved for inclusion of any research paper submitted or accepted for publication related to the MindSpace project, along with the corresponding acceptance or publication certificate.

[Attach published paper and certificate here]-**None**

B. Certificate of Competitions

This section is reserved for certificates received at technical competitions, hackathons, or project exhibitions where MindSpace was presented or demonstrated.

[Attach competition certificates here]-**None**